

A Novel Hierarchical Edge Computing Solution Based on Deep Learning for Distributed Image Recognition in IoT Systems

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Abstract—Traditionally, IoT systems utilize cloud computing platforms for ease of deployment. However, such platforms are resource-intensive, relatively expensive, and lead to longer response time due to network latency. Hence, cloud computing may not always be suitable for unusually large scale deployment. Various edge computing solutions have been proposed to overcome the above issues. However, most of such solutions rely upon expensive edge servers, which make them unsuitable for distributed applications like image recognition in agricultural fields. In this paper, we propose a hierarchical edge computing-based image recognition system in which the major processing is carried out at low-cost gateway devices like Raspberry Pi. As an example case, we address the issue of recognition of animals intruding in agricultural fields. We implement a dynamic learning method, in which a convolutional neural network is dynamically trained to recognize potential target classes based on a specific deployment environment. The AI detection module is then loaded on to the lowest level of edge servers on gateway devices for detection of animals and providing feedback. Experiments show that our proposed recognition system can perform offline image classification tasks with up to seven times higher accuracy and more than two times faster evaluation time in comparison with general-purpose cloud recognition systems. Besides, it consumes less than 6% of the network bandwidth and only a fraction of the energy as well as other computational resources compared with existing approaches.

Index Terms—Edge Computing, Image Recognition, Internet of Things

I. INTRODUCTION

Corresponding to the advance in information and operational technologies, the Internet of things (IoT) has emerged as one of the most prominent technologies in this era. In the current Internet of things paradigm, everything has computing capability and connects to the Internet. Thus, enabling conventional electronic devices such as a coffee maker, security camera, refrigerator or even toilet to generate sensor data, process

information and communicate with each other or the central server without any human intervention. [1] This concept of "machine talking to machine without direct human action" [2] has inspired countless IoT applications that proved its usefulness in many fields such as household [3], transportation [4], or agriculture [5]. For example, a security camera could detect the facial features of a criminal in an area, and alert nearby police officers or a smart refrigerator could notify family members when they need to do some grocery. With an estimated 50 billion of IoT devices deployed worldwide by 2020 [6], we might be anticipating the Internet of things revolution which affects everything from industry, economy, science, to our daily life [7].

Because of the ease of deployment and handling of the generated data, cloud computing has so far widely been used for IoT. However, many experts pointed out that cloud alone might not be an ideal solution for IoT. [8] At the moment, billions of things in IoT application generate a considerable amount of data simultaneously. Those data are usually sent to process on the cloud, which is also responsible for most decision making afterward. This approach proved to be efficient until now. However, data produced from things are expected to reach 10.4 zettabytes by 2019 [1]. Transferring that massive amount of data to the cloud would cause a massive bottleneck in that particular network and consume overwhelmingly enormous computing resources on the cloud itself. Additionally, data collected by IoT devices in households or work environment might contain people private information (e.g., individual picture or record of private conversation). Thus, raising security concerns in IoT applications.

Various edge computing solutions have been proposed to overcome the above issues. Edge computing introduces the concept of processing data at the edge of the network. Gener-

ally, the "Edge" could refer to anything capable of computation between the data source and the central cloud. [1] In edge computing paradigm, the edge, which is operating close to the data source, is responsible for most data processing and decision-making. Thus, reducing both network traffic With advance in software and hardware, Internet of things (IoT) has emerged as one of the most prominent concept. In the current Internet of things paradigm, everything has computing capability and connects to any existing network infrastructure or the Internet. Thus, enabling common electronic devices such as a coffee maker or security camera to generate sensor data, process information and communicate with each other or the central server without any human intervention. [1] This concept of "machine talking to machine without human direct action" [2] has inspired countless IoT applications that proved its usefulness in many fields such as household [3], transportation [4], or agriculture [5]. For example, a security camera could detect the facial features of a criminal in an area, and alert nearby police officers or a smart refrigerator could notify family members when they need to do some grocery. With an estimated 50 billion of IoT devices deployed worldwide by 2020 [6], we might be anticipating the Internet of things revolution which affects everything from industry, economy, science, to our daily life [7].

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Recently, researches are being carried out to use smart-phones [10] or small microcontrollers like Raspberry Pi or

Arduino. As edge servers close to the locations where data is being generated. Usually, some simple data handling or storage operations are being targeted on such edge servers.

In this paper, we addressed the issue of distributed image recognition with gateway level edge servers at the core of the process. As an example case, we address the problem of detection of animals intruding into agricultural fields. Since animal intrusion leads to considerable damage to the agricultural produce, it is essential to be able to detect animals as they enter the fields and drive them away. A traditional approach is to deploy a scarecrow (or series of scarecrows) or human labor. While animals get used to traditional scarecrows, the use of human labor is tiresome or expensive. Hence we have developed a low-cost IoT based system that would do the task automatically. At the core of the system lies the detection of the animals. For the purpose, we design and develop an Internet of things (IoT) application framework with high resource efficiency and dynamic platform scalability. We implement a highly optimized convolutional neural network (CNN) to perform specific image recognition or classification tasks on the edge of the network. In our framework, the recognition system is optimized only for a specific purpose and can dynamically adapt to the deployment environment. We name our framework as Deployment Environment Aware Learning (DEAL). To the best of our knowledge, no other literature reports implementing optimized neural networks on low-end edge devices like Raspberry Pi for image recognition applications.

Our significant contributions in this manuscript are (i) the framework of a dynamically adaptable convolutional neural network that is capable of adjusting the recognition module dynamically depending on the target deployment environment (ii) the gateway level edge server implemented on Raspberry Pi. We believe it is one of the pioneering implementations of edge server using Android Things OS after its official release on May 2018. (iii) the mechanism of detection of animals at the edge server and corresponding feedback to the higher level of the edge hierarchy, e.g., the cloud and (iv) the optimization technologies for saving resources like power consumption and network bandwidth.

Experiments show that our proposed recognition system can perform offline image classification tasks with up to seven times higher accuracy and more than two times faster evaluation time in comparison with general-purpose cloud recognition systems. Besides, it consumes less than 6 % of the network bandwidth and only a fraction of the energy as well as other computational resources compared with existing approaches.

The rest of the paper is organized as follows. In Section II, we review some of the related works that have been done previously. In Section III, we describe the design and development of our proposed recognition system. In Section IV, we evaluate our proposed scheme. We conclude in Section V.

II. RELATED WORKS

There are three major research areas related to our proposed solution.

The first is the development of the animal image recognition system. The primary challenge in this research area is to improve the efficiency and effectiveness of the recognition engine. Up until now, there have been several attempts to develop animals recognition system using computer vision and machine learning technique, each with its objective and solution to the problem. In one research paper, authors implement a pattern matching method using normalized cross-correlation to detect animal presence from a video frame [11]. While the authors of [12] proposed a method for the top view animal detection using thermal cameras for UAV technology, which implemented discrete cosine transform for feature extraction and k-nearest neighbor to classify the images. Another research dedicated to the recognition of different types of cows by employing Content-Based Image Retrieval method [13]. From our point of view, each study herein has its own specific objective and requirement that make each solution criteria of functional efficiency and effectiveness varied. Our proposed method aims to improve detection reliability, response time, and resource efficiency, which is crucial for on-device-animal-detection-system that operate mainly in agricultural fields.

Another related research field is the study of the deep neural network for image classification. Since the pioneering publication of LeNet-5 [14] and AlexNet, [15] outstanding performance in ImageNet [16] Large Scale Visual Recognition Challenge in 2012, convolutional neural network (CNN) has been ubiquitous in solving complex computer vision challenge. The major effort in this area, including attempts to design a network that can solve more complex image recognition task with higher accuracy and studies that aim to improve network efficiency by various algorithms. Several CNN based models have created important milestones in this research areas such as ZFNet [17], VGG net [18], GoogLeNet [19], Microsoft ResNet [20], R-CNN [21], Fast R-CNN [22], Faster R-CNN [23]. Our proposed model is inspired and based on MobileNets [24], which aim to provide efficient CNN for mobile and embedded devices.

The last related area of study is the application of edge computing in the Internet of things application. Edge computing [25] [26], also known as Fog Computing [27], and Cloudlet Computing [28], is still a relatively new area of research that aims to distribute the computing tasks to the edge of the network. As cloud-based computing paradigm becomes less practical for large-scale IoT application, edge computing rises to resolve the challenge of IoT application deployment. The authors of [1] refer to "Edge" as any computing capable devices between the data source and the cloud server. Conventional edge devices we found in relevant studies have included smartphones [10], smart wearable devices, general computer servers, and embedded devices. Our proposed method deploys recognition engine on embedded edge devices such as Raspberry Pi to decentralize computing process from the cloud

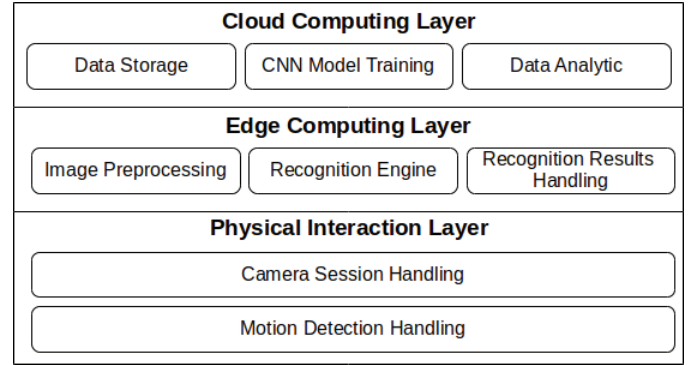


Fig. 1. IoT Kakashi hierarchical system architecture

server.

III. IOT KAKASHI: A HIERARCHICAL EDGE COMPUTING SOLUTION FOR ANIMAL IMAGE RECOGNITION

Traditionally, cloud computing platforms with high computational capability handle most complex computing tasks, including image recognition. Nowadays, however, the traditional approach is not suitable anymore, considering the massive amount of IoT devices that are to be connected to the cloud. In our proposed system, the recognition engine is deployed on the edge device to optimize the execution time. Using a combination of the Internet of things, deep learning and edge computing techniques, we develop an edge-based distributed animal recognition system that can detect animals presence and determine the types of animals.

A. System Architecture

Inspired by an edge computing framework called "Dynamic Hierarchical Edge Architecture" (DHEA) [9], our animal recognition system consists of three operational layers called Physical Interaction Layer, Edge Computing Layer, and Cloud Computing Layer. Fig. 1 illustrates our system architecture. The Physical Interaction Layer handles tasks related to real-world data, including data acquisition, motion detection, capturing animal images, or physical feedback. Both sensors and actuators belong to this layer. The Edge Computing Layer is where core computation processes take place. This layer handles high-frequency computing tasks, including image processing, animal recognition, data processing, temporary data storage, and also serve as a communication channel between Physical Interaction Layer and Cloud Computing Layer. IoT devices in our system belong to this layer, and thus called edge devices. For animals detection in the agricultural fields, such edge servers are deployed at different locations of the fields. Each edge server can function independently and may have a different set of animals to be detected. Hence, the task of detection of animals is distributed among the edge servers. The Cloud Computing Layer handles occasional but complex tasks such as training CNN and data analytic. This layer also serves as a storage for a large amount of training data that require preprocessing. High-performance cloud servers belong to this

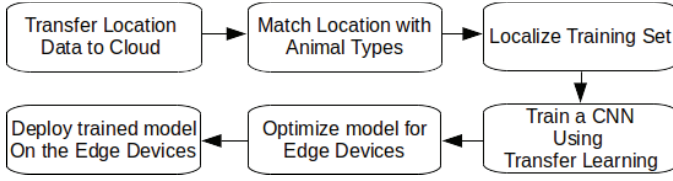


Fig. 2. Deployment Environment Aware Learning Process for CNN based model

layer. The Edge Computing Layer and Cloud Computing Layer often exchange information. However, the only information that can be used in a machine learning algorithm or data analysis is transmitted to the Cloud Computing Layer.

B. Deployment Environment Aware Learning (DEAL)

In this section, we described our proposed "Deployment Environment Aware Learning (DEAL)" framework. Our recognition engine is based on a MobileNets convolutional neural network [24]. We use Tensorflow open-source library to train CNN, and the results are in Tensorflow Lite format.

On, Cloud Computing Layer, we train a new optimized CNN based on pre-trained MobileNets using transfer learning technique. This approach minimizes training time and amount of training data required, as well as encourages rapid continuous improvement of the recognition engine after deployment. As shown in Fig. 2, the overall training process in DEAL framework are the following.

- 1) Deployed edge devices generate location data using GPS on Physical Interaction Layer and transmit that data to Cloud Computing Layer via the Internet.
- 2) On Cloud Computing Layers, transmitted location data are processed and compared with animals database. The results are a list of potential animals that can be found at the target location.
- 3) The list of potential animals is used to generate unique training data by pulling image data from an animal image database and fetch more if required from the Internet with web crawlers. The results are training data consisting of images from animals that frequent the area.
- 4) The generated training data is used, together with the based CNN model, in the transfer learning algorithm. The result is a unique model built specifically for detecting animals in the target environment.
- 5) The trained model is optimized by compressing algorithm to make it suitable for deploying on edge devices over the Internet. The result is relatively small, and fast CNN optimized for particular edge devices.
- 6) The optimized model is distributed to all edge devices in the target location. Thus, conclude the training process in DEAL framework.

Cloud-based image recognition services, e.g., Google Cloud Vision, Amazon Rekognition, and Clarifai usually aim to provide a universal general-purpose image recognition engine trained to recognize countless many classes spanning across a vast number of categories. Such recognition engines are

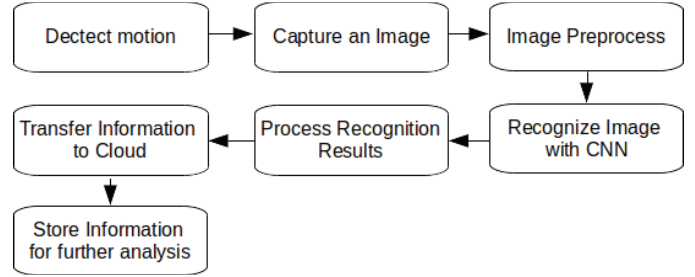


Fig. 3. Animal Recognition Process Overview

TABLE I
EDGE DEVICE INFORMATION

Raspberry Pi 3 Model B	
CPU	1.2 GHZ quad-core ARM Cortex A53
GPU	Broadcom VideoCore IV @ 400 MHz
Memory	1 GB LPDDR2-900 SDRAM
Network	10/100 MBPS Ethernet 802.11n Wireless LAN
OS	Android Things 1.0

suitable for general image recognition on a wide range of categories but tend to confuse and become unreliable when dealing with complicated images. Our recognition engine takes an entirely different approach by minimizing the number of target classes based on the requirement of the location (or environment). Using our training method called "Deployment Environment Aware Learning," we trained our recognition system with classes of animals commonly spotted around the particular deployment environment.

C. Animal Recognition Process

Fig. 3 illustrates the flow of our recognition process, which encompasses every layer in our system architecture. The overall animal recognition process is described as the following.

- 1) The recognition process is triggered by a motion-sensing module on the Physical Interaction Layer that sends signals to the edge device on Edge Computing Layer.
- 2) The edge device then initiates camera handling session in collaboration with a camera module on Physical Interaction Layer to capture animal images.
- 3) The edge device pre-processes captured image which is initially a bitmap by re-scaling it to 224x224 resolution and saves it in PNG format
- 4) The recognition engine embedded in the edge device then performs image recognition task locally using the saved PNG image as an input. Reducing the size of the input image beforehand reduces the computing resource consumed by CNN as well as recognition process time.
- 5) The results generated from CNN are processed into a more insightful format and portable formats such as JSON and XML.
- 6) The converted results are sent to the cloud server on Cloud Computing Layer for further analysis. When accumulated, these results can be used to improve our model in the future.

IV. PERFORMANCE EVALUATION

We developed a proof-of-concept system to evaluate the quality of IoT application using our proposed framework. For this experiment, we deployed our animal recognition system in an agricultural region of Tsuruoka city in Yamagata prefecture of Japan. In recent years, several farmers in Tsuruoka have reported losing their crops to raiding wild animals. From our observation, common animal types spotted in this area are monkeys, birds, bears, and wild boars. As such, these types of animals were used as target animal classes for DEAL enhanced training process. For this research, our primary objectives are to provide fast response time and reliable detection result. Accordingly, if there is a monkey in the area, our recognition system should report a monkey presence as soon as it enters the motion detection range.

Table I describes edge device system information. Due to its affordable price and high computing capability, we chose the Raspberry Pi 3 Model B as our prototype development platform. For motion detection, we employed a PIR sensor that detects infrared (IR) light radiating from objects in its field of view. Pi Camera is used to capture animal images. Lastly, we use Android Things OS as the operating system. Fig. 4 shows an overview of our developed system.

We compare the performance of our system with that of cloud-based image recognition services, including Google Cloud Vision, Amazon Rekognition, and Clarifai. To evaluate the performance of each of these engines, we run the same recognition tasks on every recognition system. Afterward, we evaluate our model performance Top-1 and Top-3 accuracy. [15].

In animal recognition system, high accuracy and fast response time are crucial because users always expect highly accurate results, and they also want to respond to animal intrusion as soon as possible.

As observed in Table II, our proposed recognition engine, which has been trained using Deployment Environment Aware Learning (DEAL), shows up to seven times higher accuracy than other mechanisms in Top-1 accuracy category. It should be noted that since the correct detection result would allow the system to develop high precision animal repelling technique, Top-1 accuracy is essential for this animal recognition use case. Also, our proposed scheme scores higher in Top-3 accuracy in comparison with other general-purpose cloud-based solution for animal image recognition.

Tables III and IV show the evaluation time and execution time taken by each system under comparison. It can be observed that our proposed scheme leads to more than two times faster minimum evaluation time and more than two times faster minimum execution time.

Table V shows the network bandwidth used by each system in comparison. It can be observed that our proposed system uses less than 6 % of network bandwidth compared with other existing schemes. This is important for a system that needs to be deployed in the field where internet connection might be slow.

Finally, we conducted a long term experiment to estimate our system energy consumption. The experiment setup consists of our prototype system (a Raspberry Pi 3, a PIR sensor module, and a Pi Camera) and a portable 3g router, both of which are powered by a 12v 45AH battery. The recognition engine is configured to simulate a recognition process, as shown in Fig. 3, every 5 minutes. As a result, our prototype lasts averagely 7 days before the battery run out of power. Further experiment with a solar panel attached to the prototype will be carried out in the future.

TABLE II
ACCURACY RATE OF OUR PROPOSED SYSTEM COMPARED TO CLOUD-BASED SOLUTION

Model	Top-1 accuracy	Top-3 accuracy
Our Proposed System	0.9	0.96
Google Cloud Vision	0.57	0.77
Amazon Rekognition	0	0.7
Clarifai	0.13	0.2

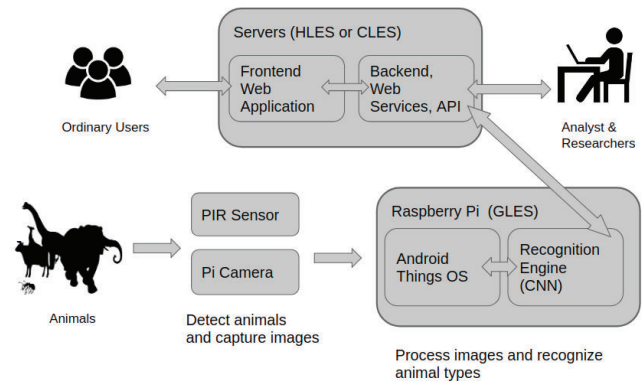


Fig. 4. System overview

TABLE III
EVALUATION TIME OF OUR PROPOSED SYSTEM COMPARED TO CLOUD-BASED SYSTEM

Model	Min Eval. Time	Max Eval. Time
Our System	0.3190	1.6400
Google Vision	0.7233	1.3768
Amazon Rekognition	1.2835	2.5770
Clarifai	0.9491	1.8757

TABLE IV
EXECUTION TIME OF OUR PROPOSED SYSTEM COMPARED TO CLOUD-BASED SYSTEM

Model	Min Exec. Time	Max Exec. Time
Our System	0.962	3.039
Google Vision	2.456	12.571
Amazon Rekognition	2.596	8.446
Clarifai	2.147	8.803

V. CONCLUSION

In this paper, we propose an edge-based Internet of things solution to detect the intrusion of wild animals in agricultural

TABLE V
BANDWIDTH USAGE OF OUR PROPOSED SYSTEM TO CLOUD-BASED SOLUTION

Model	Bandwidth Usage (kb/s)
Our Proposed System	4.3
Google Cloud Vision	71.8
Amazon Rekognition	70.7
Clarifai	72.5

fields. We propose an efficient and effective animal recognition system based on a convolution neural network model to detect a specific type of animals in the deployment area. While cloud-based recognition service such as Google Vision, Amazon Rekognition, or Clarifai provide a universal general-purpose model, we propose an optimized and specialized model trained using Deployment Environment Aware Learning (DEAL) method. With DEAL, the model is trained based on specific types of animal populated in the target deployment area. Even though we use a high-performance cloud server for the training process, we deploy our recognition engine directly on the relatively low performance and power-efficient edge devices to perform offline recognition tasks. This approach reduces computation on the cloud, which allows for large-scale deployment in the future and efficiently improves recognition performance. Experiments have shown that our proposed recognition system can perform offline image classification tasks with up to seven times high accuracy and more than two times faster evaluation time in comparison with general-purpose cloud recognition systems. Also, it consumes less than 6% of the network bandwidth and only a fraction of the energy as well as other computational resources compared with existing approaches.

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